

Aethon

The Autonomous Enterprise Asset Memory & Intelligence Engine

1. Executive Summary

In modern industrial and enterprise environments, critical data is scattered across disconnected silos: maintenance logs, compliance manuals, sensor streams (IoT), work orders, and incident reports. When an asset fails or a compliance audit occurs, engineers and operators waste countless hours manually correlating this fragmented information.

Aethon is an AI-powered Unified Asset Memory Engine. It ingests, interconnects, and analyzes enterprise data to create a centralized Knowledge Graph. Using a collaborative multi-agent AI architecture, Aethon acts as an autonomous Subject Matter Expert (SME), enabling predictive maintenance, instant root cause analysis (RCA), and automated compliance auditing.

2. The Problem

- **Data Silos:** Sensor data lives in SCADA systems, manuals are PDFs on SharePoint, and work orders are in ERPs.
 - **Reactive Maintenance:** Unplanned downtime costs industrial facilities millions. Transitioning from reactive to predictive maintenance requires deep data correlation.
 - **Compliance Risks:** Manual auditing of Standard Operating Procedures (SOPs) against live asset conditions is slow and error-prone.
 - **Knowledge Loss:** When experienced engineers retire, their undocumented troubleshooting knowledge leaves with them.
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3. The Aethon Solution

Aethon solves these challenges by treating the enterprise facility as a living, interconnected graph, continuously monitored and analyzed by specialized AI agents.

Core Capabilities

1. **Unified Asset Knowledge Graph** Aethon extracts entities and relationships from uploaded documents (PDFs, CSVs, Manuals) and maps them to physical assets. This graph interconnects Assets, Maintenance Logs, Sensors, Incidents, and Vendors into a single queryable memory bank.
2. **Autonomous Root Cause Analysis (RCA)** When an asset degrades, Aethon's Diagnostic Agent traverses the Knowledge Graph, correlating historical incidents, recent sensor anomalies, and manufacturer guidelines to instantly deduce the root cause and recommend fixes.
3. **Predictive Maintenance (PdM) & Health Forecasting** Aethon aggregates real-time metrics and historical failure patterns to forecast asset degradation before it happens, automatically generating preemptive work orders.

4. **AI-Driven Compliance Auditing** Aethon constantly evaluates live facility data against regulatory documents and internal SOPs. When a violation is detected (e.g., missed safety inspections), Aethon flags the conflict and auto-generates a compliant rewrite or action plan.
 5. **Multi-Agent AI Collaboration** Aethon is not a simple chatbot. It utilizes a network of specialized agents (Data Ingestion, Diagnostic, Compliance, and Workflow Agents) that collaborate to solve complex, multi-step engineering tasks autonomously.
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4. Technical Architecture

Aethon is built on a modern, scalable, and highly performant architecture, designed for rapid deployment and enterprise reliability.

Frontend (User Interface)

- **Framework:** Next.js (React) with Server-Side Rendering (SSR) for optimal performance.
- **Styling & Animation:** Tailwind CSS and Framer Motion for a premium, highly responsive, and dynamic user experience.
- **Visualization:** Interactive Canvas for real-time Knowledge Graph traversal.
- **Hosting:** Deployed on **Cloudflare Pages** for global edge delivery.

Backend (API & AI Orchestration)

- **Framework:** FastAPI (Python) for asynchronous, high-throughput API endpoints.
- **AI Integration:** OpenAI (GPT-4o) and LangChain for multi-agent reasoning, RAG (Retrieval-Augmented Generation), and NLP.
- **Vector Database:** ChromaDB / Pinecone for semantic embedding storage and fast vector search.
- **Relational Database:** PostgreSQL for structured asset tracking, sensor logs, and user metadata.
- **Hosting:** Containerized via Docker and deployed on **Render**.

Data Flow

1. **Ingestion:** Documents are uploaded, chunked, and embedded. Entities are extracted via LLM to build graph relationships.
 2. **Reasoning:** User queries trigger the Copilot Router, which delegates tasks to the appropriate specialized Agent.
 3. **Synthesis:** The Agents query the Vector DB and SQL DB, synthesize an answer, and stream it back to the client via WebSockets.
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5. Key Differentiators

- **Context over Search:** Instead of just doing keyword searches, Aethon *understands* the physical relationships between a broken pump, its upstream valve, and the maintenance manual.
 - **Action-Oriented:** Aethon doesn't just answer questions; it generates executable Work Orders, Shift Reports, and Emergency Plans.
 - **Stunning UI/UX:** Designed with a "dark mode glassmorphism" aesthetic that turns complex industrial data into a beautiful, intuitive digital twin dashboard.
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6. Business Impact

- **Reduced Downtime:** By forecasting failures and accelerating RCA, Aethon minimizes unplanned facility outages.
 - **Operational Efficiency:** Automates the generation of shift reports and work orders, freeing up engineers for high-value tasks.
 - **Risk Mitigation:** Continuous automated compliance audits prevent costly regulatory fines and safety incidents.
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Built with ❤️ for the Hackathon.